

# Classifying Emergence by Information-Flow Operations on the Path Landscape

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## Abstract

Emergence is ubiquitous across physics, biology, neuroscience, and artificial intelligence, yet existing classifications remain either substrate-specific or restricted to coarse weak / strong dichotomies. We introduce the *path landscape* — a representation of a system’s information flow as the distribution of input-to-output paths through its (possibly recurrent, multi-scale) connectivity, grouped into similarity modes. Within this representation, emergence is not a property of any single path; it is the visible signature of how the *distribution* of paths is organized between fixed inputs and fixed outputs. We argue that every emergent phenomenon can be read as a combination of four primitive operations on this distribution: **path activation**, **path suppression**, **mode merge**, and **mode split**. These four operations form a  $2 \times 2$  algebra (paths vs. groups  $\times$  creation/fusion vs. removal/fission). Each produces a qualitatively distinct kind of emergence at the output — the arrival of new capability, selective focusing, feature collapse / global integration, and feature differentiation, respectively. We illustrate the framework with ten emergent phenomena drawn from ferromagnetism, Bénard convection, speciation, cell differentiation, flocking, categorical perception, working memory, edge detection, grokking in neural networks, and pidgin-to-creole language formation, showing that the classification covers diverse fields and accommodates natural mixtures.

## 1 Introduction

Emergence — the appearance, at some macroscopic level of a system, of properties that are not visible at the level of its parts — is ubiquitous, but it has resisted a mechanism-grounded classification. Existing taxonomies either partition emergence by substrate (“physical”, “biological”, “cognitive”) or by epistemic strength (“weak” vs. “strong”), neither of which exposes *how* the underlying machinery actually produces the emergent output.

In this paper we approach the problem from the side of *information flow*. Given a system with identifiable inputs and identifiable outputs, we represent the system by the distribution of routes that information takes from one to the other. We call this the *path landscape*. We then ask: what are the qualitatively distinct ways in which this distribution can be organized so that the output looks emergent?

The central claim of the paper is that there are exactly four: information-processing circuits can come online (**activation**), existing circuits can be silenced (**suppression**), groups of similar routes can fuse (**merge**), or one group can fission into several (**split**). Each operation has a clean structural fingerprint and a recognizable emergent signature. Together they form a small algebra in which we believe every emergent phenomenon can be located, either as a dominant operation or as a mixture.

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## 2 The path landscape

### 2.1 Definition

Consider any system that can be cast as a directed graph  $G = (V, E)$  of *computing units*  $V$  connected by *interactions*  $E$ . Each interaction carries a (possibly continuous) weight  $w_{uv} \in \mathbb{R}$ , and may be feed-forward or recurrent. Distinguish a set of input units  $V_{\text{in}} \subset V$  and a set of output units  $V_{\text{out}} \subset V$ . The remaining units are internal.

If  $G$  contains recurrent edges, unroll it in time over  $T$  discrete steps, producing the static DAG  $G^{(T)}$  in which each unit  $u$  becomes a copy  $u@t$  for  $t = 1, \dots, T$ , and each recurrent edge  $u \rightarrow v$  becomes a forward-only edge  $u@t \rightarrow v@(t+1)$ . The unrolled graph  $G^{(T)}$  has no loops by construction.

A *path* from input to output is an ordered sequence of nodes

$$\pi = (s_1, s_2, \dots, s_L), \quad s_1 \in V_{\text{in}}, \quad s_L \in V_{\text{out}},$$

with each  $(s_i, s_{i+1})$  an edge of  $G^{(T)}$ . We assign to  $\pi$  a non-negative weight  $\omega(\pi)$  that combines the magnitudes of the edges it traverses (e.g. a product or a normalised flow). Let  $\Pi$  be the set of all such paths.

### 2.2 Routes and modes

Many paths in  $\Pi$  differ only in time-step bookkeeping ( $u@t$  vs.  $u@t'$ ) or in a brief lingering on a self-loop. Stripping these gives a *route* — the base sequence of distinct units. Let  $R$  be the set of routes. Each route  $r \in R$  inherits the cumulative weight of the paths that collapsed onto it.

Two routes are *similar* if they share many of the same units and edges; for instance, a composite of edge-Jaccard overlap and an ordered longest-common-subsequence measure. Clustering  $R$  under this similarity produces *modes*: equivalence classes of routes that compute essentially the same function. The set of modes  $M = \{m_1, m_2, \dots, m_k\}$ , together with the route weights, is the *path landscape*:

$$\mathcal{L}(G, V_{\text{in}}, V_{\text{out}}) = (R, \{\omega(r)\}_{r \in R}, M).$$

Three numbers summarize the landscape at a glance: the number of routes  $|R|$ , the number of modes  $|M|$ , and the rank-distribution of total mode weight.

### 2.3 What the landscape sees

This representation is structural, but it absorbs much of the continuous dynamics one might fear it discards. Weight magnitudes, synaptic gains, and effective couplings are encoded in  $\omega(r)$ . Coherence, phase relationships, and dynamical co-activation are encoded in the similarity structure that defines  $M$ : paths that fire together over time become close in the similarity metric and collapse into the same mode. A continuous attractor, for example, appears as a band of nearly-identical high-weight routes whose similarity matrix is approximately block-rank-one. Gain modulation appears as a re-shaping of the weight rank-distribution without a change of topology.

### 2.4 Examples of systems

- A *neural circuit* — units are neurons or micro-circuits, interactions are synaptic connections, recurrence is structural feedback, inputs are sensory afferents, outputs are motor or downstream targets.

- *A deep neural network* — units are channels or attention heads, interactions are weight matrices, inputs are the embedding layer, outputs are the prediction head.
- *A biochemical regulatory network* — units are genes or metabolites, interactions are regulatory edges, inputs are external signals, outputs are phenotypic markers.
- *A flock* — units are birds, interactions are near-neighbor coupling, inputs are environmental cues, outputs are collective motion.
- *A magnetic lattice* — units are spins, interactions are exchange couplings, “input” is the boundary condition or external field, “output” is the macroscopic magnetization.

In each case the same definition applies; the differences across fields are reduced to differences in the structure and weighting of  $\mathcal{L}$ .

### 3 Emergence as transformations of the path distribution

If a system is fully described by  $\mathcal{L}$ , then anything that deserves to be called emergence must be a property of how  $\mathcal{L}$  is organized. We define emergence operationally as a change in  $\mathcal{L}$  that produces a recognizable signature at the outputs  $V_{\text{out}}$ .

The transformations of  $\mathcal{L}$  have a small irreducible set of primitives. They are organized along two binary axes (Table 1): where the change acts (on individual paths, or on groups of paths — modes), and the direction of the change (creation/fusion versus removal/fission).

|               | create / fuse        | remove / fission      |
|---------------|----------------------|-----------------------|
| <b>paths</b>  | 1. <b>Activation</b> | 2. <b>Suppression</b> |
| <b>groups</b> | 3. <b>Mode merge</b> | 4. <b>Mode split</b>  |

Table 1: The four primitive operations on the path landscape.

We discuss each in turn, noting both its formal effect on  $\mathcal{L}$  and the emergent signature it produces at the output. A geographical metaphor, with information flow imagined as a river shaped by the landscape’s geometry, is used throughout to keep the intuition concrete.

#### 3.1 Path activation: a new channel opens

A new route  $r \notin R$  becomes viable, either because a previously silent edge becomes effective or because deeper unrolling reveals a longer path that traverses a loop. Formally,  $R \rightarrow R \cup \{r\}$ .

The emergent signature is the *appearance of a new capability* at the output — a new behavior, a new representation, a new qualitative state. The output now does something it could not do before, because there is now a route along which information of the required kind can flow.

*Geography.* A new river course is carved: the rain has the same total volume, but a new path through the terrain becomes viable, and the lake at the bottom of the basin now receives water it did not previously receive.

*Recurrent loops as a sub-case.* Recurrence does not require a new “time-emergence” category. In the unrolled graph, a path that revisits a unit via a feedback edge is simply a longer path, and it becomes available at a sufficient unrolling depth. The emergent signature — persistence, attractor convergence, iterative refinement — is then exactly the signature of activation, applied to a long re-entrant route.

### 3.2 Path suppression: a channel is dammed

An existing route  $r \in R$  is silenced: either because of inhibition along one of its edges, or because of gating that prevents weight from flowing along it. Formally,  $R \rightarrow R \setminus \{r\}$ , or equivalently  $\omega(r) \rightarrow 0$ .

The emergent signature is *selective focusing*: the output becomes sharper not because anything has been added but because competing flow has been removed. Lateral inhibition produces edge-like features by suppressing paths that carry the homogeneous background. Attention picks out a target by suppressing distractor routes. Signal emerges from noise because noise paths have been silenced.

*Geography.* A dam is built across a channel: the water that used to dissipate sideways is now confined to the remaining route, which becomes recognizable as a sharp downstream feature.

### 3.3 Mode merge: a gorge concentrates flow

Two (or more) modes  $m_i, m_j \in M$  fuse into a single mode  $m_{ij}$ . Formally,  $M \rightarrow (M \setminus \{m_i, m_j\}) \cup \{m_{ij}\}$  with  $\omega(m_{ij}) = \omega(m_i) + \omega(m_j)$ .

The emergent signature is *feature collapse and global integration*: paths that previously computed distinguishable sub-features now collapse onto a shared output sub-pattern. Categorical perception, binding, abstraction, holistic representation, generalization, and the formation of macroscopic order from microscopic configurations all live here.

*Geography.* The terrain narrows into a gorge: many tributaries that were running roughly parallel are forced into a single river, and the downstream observer no longer sees the tributaries individually — only the concentrated current.

### 3.4 Mode split: a delta spreads flow

A single mode  $m \in M$  fissions into two or more sub-modes  $m_a, m_b, \dots$ . Formally,  $M \rightarrow (M \setminus \{m\}) \cup \{m_a, m_b, \dots\}$  with  $\omega(m_a) + \omega(m_b) + \dots = \omega(m)$ .

The emergent signature is *feature differentiation*: a previously uniform output space acquires distinguishable sub-patterns. Specialization, discrimination, symmetry-breaking, expert tuning, and the appearance of novel sub-categories all live here.

*Geography.* The river enters a delta: a single channel fans out into distinguishable sub-channels, and downstream the flow is recognized as a structured set of distinct streams rather than one uniform current.

### 3.5 Composition and mixtures

Real emergent phenomena are rarely pure. Categorical perception is mostly mode merge (within-category collapse) but contains a thin admixture of mode split (between-category sharpening). Grokking is mostly path activation but is followed by a merge in which the new circuit unifies many examples. Speciation is split in its dominant operation but is enabled by suppression of inter-population gene flow. The classification is not a partition; it is a basis. Every emergent phenomenon decomposes into a (possibly time-extended) combination of the four operations.

## 4 Schematic of the four operations

Figure 1 sketches the four operations in side-by-side, low-information form: each panel shows a small input fan, a few routes through internal units, and a small output fan, and a thumbnail of the river geography that motivates the name.

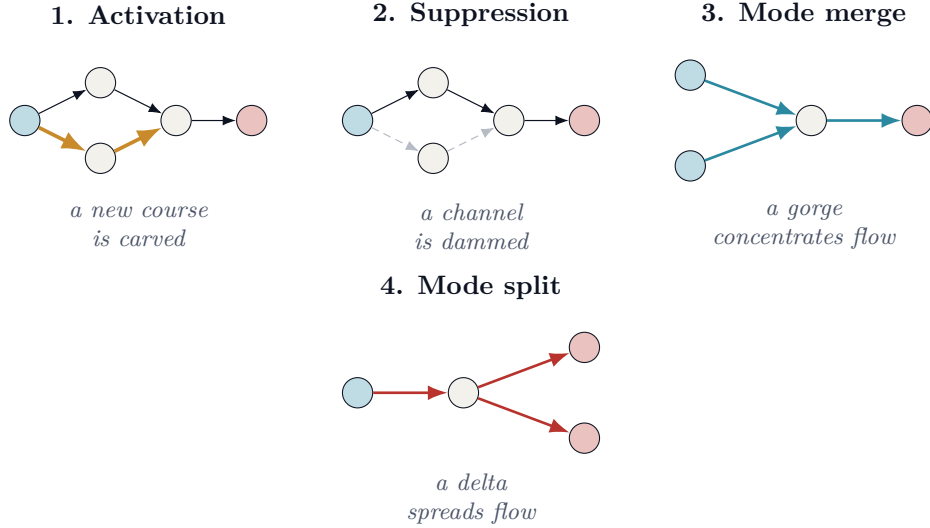


Figure 1: Schematic of the four primitive operations on the path landscape. Cyan circles are input units; red circles are output units. **Activation** introduces a new route (amber). **Suppression** silences an existing route (dashed grey). **Mode merge** concentrates routes into a shared sub-pattern. **Mode split** fissions one sub-pattern into several. Each panel is captioned with its river-geography metaphor.

## 5 Causes: the geography of information flow

A useful way to think about *why* a system exhibits one operation rather than another is to remember that, in the river metaphor, the operations are not chosen by the water — they are imposed by the geography. The path landscape *is* the geography of the system, and the four operations correspond to four canonical shapes that the landscape can take.

**Why activation occurs.** The system has acquired a new viable course. In neuroscience this might be the maturation of an axon tract, in machine learning a circuit that emerged from gradient descent, in linguistics a regularization performed by child learners. In each case the cause is a structural change in the graph  $G$ , or a deepening of the unrolling  $T$ , that makes a new route from input to output possible. *Cause: new connectivity, or deeper recurrence.*

**Why suppression occurs.** The system has installed a gating mechanism that selectively damps existing routes. In visual cortex this is lateral inhibition; in attention it is top-down gain control; in evolutionary biology it is reproductive isolation. *Cause: inhibitory or gating connectivity that operates on routes already present in  $\mathcal{L}$ .*

**Why mode merge occurs.** Multiple routes converge in their structure or in their effective output — either because they share intermediate units, because their endpoints coincide, or because their dynamical activity becomes correlated. In physics, falling below a critical temperature aligns spins; in perception, a learning history of equivalent reinforcement collapses categories. *Cause: shared structure or shared weighting that forces formerly distinct modes into one.*

**Why mode split occurs.** A previously uniform mode is broken apart by a structural or weighting difference that the landscape is now sensitive to — a symmetry-breaking bifurcation, a developmental signal, a learning experience that introduces a relevant contrast. *Cause: a structural or weighting asymmetry that fissions a previously homogeneous mode.*

In all four cases the cause is the same in form: the geometry of  $G$  (together with the unrolling and the weighting) makes a particular operation on  $\mathcal{L}$  inevitable. The cause of emergence is the cause of the operation that produces it.

## 6 Case studies across fields

To illustrate the classification we discuss ten emergent phenomena drawn from physics, biology, neuroscience, cognitive science, machine learning, and linguistics. They are grouped by the dominant operation of the corresponding path-landscape transformation.

### 6.1 Path activation (new capability appears)

**Grokking in neural networks (machine learning).** After many training steps with apparent overfitting, the model abruptly generalizes; mechanistic interpretability has identified that a new internal circuit (for example a Fourier-decomposition circuit for modular arithmetic) suddenly carries the computation. A previously dormant route through the weights takes over: this is path activation in the cleanest sense.

**Pidgin-to-creole language formation (linguistics).** Children exposed to a structurally impoverished pidgin spontaneously produce a grammatically rich creole, with recursion, tense, and case-marking that were not present in the input. New combinatorial routes from intended meaning to spoken sentence come online that the parent system did not contain.

**Working memory persistent activity (neuroscience).** A stimulus that has physically vanished is held active in prefrontal circuits for seconds. In the path landscape this is the recurrent sub-case of activation: longer paths that loop back through the same units become viable with deeper unrolling, and emergence appears as sustained re-processing of an absent input.

### 6.2 Mode merge (global integration)

**Ferromagnetism below the Curie point (physics).** Above  $T_c$ , individual spin-orientation paths are independent; below  $T_c$ , the route distribution collapses onto one dominant aligned mode, and macroscopic magnetization appears as the integration of microscopic routes.

**Categorical perception (cognitive neuroscience).** A physical continuum (color hue, the /ba/–/pa/ acoustic continuum) is heard or seen as discrete categories with sharp boundaries. Many input routes (small stimulus variations) merge onto the same output mode: the percept is the merge.

**Starling murmurations (collective animal behavior).** The flight-decision routes of thousands of individual birds become correlated across the flock. The route distribution acquires a single coherent collective mode (the flock’s direction) that no single bird’s wiring contains.

### 6.3 Mode split (feature differentiation)

**Speciation (evolutionary biology).** One ancestral gene-flow cluster fissions into two reproductively isolated populations, each carving out its own route through morphological and behavioral space.

**Cell differentiation (developmental biology).** A pluripotent stem cell, whose gene-regulatory routes form a single cluster, fissions into many specialized cell-type clusters — neurons, hepatocytes, myocytes — each tracing a distinct regulatory route through the same genome.

**Bénard convection cells (nonlinear physics).** A uniform fluid heated from below has homogeneous flow paths; above a critical temperature gradient the route distribution fissions into ordered convection cells with distinct upflow and downflow lanes. Symmetry-breaking is, structurally, mode split.

## 6.4 Path suppression (selective focusing)

**Edge detection in V1 by lateral inhibition (vision).** Center-surround receptive fields actively suppress paths from neighboring photoreceptors at regions of uniform luminance, leaving only paths at luminance discontinuities active. Edges appear not because edge-routes were added but because background-routes were silenced.

A quick reading of the distribution: physics and developmental phenomena tend to live in split (symmetry-breaking, differentiation); collective and perceptual phenomena tend to live in merge (binding, alignment, categorization); novelty-and-skill phenomena live in activation (new circuits, persistent activity); attention-and-feature-detection phenomena live in suppression. Most real cases are dominated by one operation but contain a thin admixture of the others, which is exactly what a four-operation algebra should predict.

# 7 Discussion

## 7.1 What the framework gains

Three things follow from treating emergence as transformation of the path distribution. First, the classification is mechanism-grounded: an instance of emergence is named by the operation that produced it, not by the substrate in which it lives. Second, the framework is field-agnostic — once a system can be written as a directed graph with inputs and outputs, its emergent behavior is comparable to that of any other such system. Third, recurrence and time fold into the framework without a separate axis: a recurrent loop is just a longer path that becomes active at a sufficient unrolling depth, so persistence and oscillation are sub-cases of activation rather than a parallel taxonomy.

## 7.2 Limits of the structural view

The path landscape is a projection. It absorbs continuous dynamics through path weights and inter-route similarity, but phenomena whose essence lives in the magnitude of a single signal or in the phase of a continuous oscillation are visible only as a coarse shadow. The framework is therefore most informative for emergent phenomena whose primary content is the *organization* of information flow, and least informative for those whose primary content is the raw amplitude or high-resolution phase of the flow.

## 7.3 Predictions and falsifiability

Because the operations are defined structurally, they make testable predictions. For example: if a system's emergence is classified as merge, a targeted lesion of any of the intermediate units shared across the merging routes should abolish the emergent property; if it is classified as suppression, restoring the silenced routes (by lifting the inhibition) should make the emergent feature disappear; if it is classified as activation, the operation can be turned on and off by varying the unrolling depth or the connectivity that creates the new route. The framework



is therefore not a relabeling exercise: it commits to specific causal predictions about which structural interventions will and will not destroy a given instance of emergence.

## 8 Conclusion

We have argued that emergence, despite its apparent diversity across fields, factors into a small set of operations on a single representation: the path landscape. The four primitive operations — activation, suppression, merge, and split — exhaust the qualitatively distinct ways that the distribution of input-to-output routes can be transformed, and each produces a recognizable kind of emergent signature at the output. Real phenomena are typically dominated by one operation and accompanied by minor admixtures of the others. The classification is field-agnostic, structurally grounded, and yields concrete predictions about which interventions will preserve and which will destroy any given instance of emergence. We hope it provides a useful common language across the many disciplines in which emergence is named but rarely mechanistically classified.

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